

BioPrint: behavior-based login security

A privacy-first, browser-local behavioral biometric gate that complements passwords with the signals people naturally create while typing and moving a cursor.

Problem Stolen credentials can be correct but still belong to the wrong person.	Approach Compare live interaction rhythm with a local, enrolled baseline.	Outcome An explainable decision with no network inference latency.
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Signal design

BioPrint turns low-friction browser events into aggregate behavioral measurements. Keystroke dwell time captures key press duration; flight time captures the release-to-next-press interval and its variation. Pointer movement contributes curvature (travel distance relative to direct distance) and peak velocity. Focus transitions and submit-button hold duration provide small supporting signals. Raw event streams are discarded after aggregation, and the calibration phrase itself is not retained as a biometric feature.

Architecture and classifier

During three enrollment passes, the client stores the mean and standard deviation for each signal in browser local storage. At login, it normalizes each live value with a Z-score, $|live - mean| / (standard\ deviation + tolerance)$, and combines weighted deviations into a 0-100 confidence score. A score of 70 or above is a behavioral match for this demonstration. The model is fixed-size arithmetic, so evaluation occurs locally in far below the 10 ms target.

The design is intentionally transparent and sample-efficient. With larger consented enrollment sets, the same interface can evolve to Mahalanobis distance, which captures correlation among signals such as fast typing and rapid pointer movement.

System flow

1. ENROLL Three natural passes establish a local baseline.	2. COLLECT A new login produces aggregate timing and motion features.	3. DECIDE Bot rules run before local anomaly scoring.	4. EXPLAIN Dashboard exposes confidence and the top mismatch.
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Security controls that judges can see

The demo separates an automated attack from an ordinary behavioral mismatch and makes the decision legible in real time.

Anti-bot and anti-replay controls

Before profile matching, BioPrint flags three suspicious patterns: untrusted browser events, near-zero variation across at least three key-flight times, and a perfectly linear observed pointer path. The last two should be treated as risk indicators rather than proof; browsers and assistive technologies vary. The UI reports a distinct orange bot state, while a low-confidence genuine-browser attempt receives a red mismatch state.

Privacy and responsible deployment

The prototype makes the most privacy-preserving architectural choice for a hackathon: all processing and profile storage occur in the browser. No behavioral event data is sent to an API. A production deployment should encrypt profile material, provide consent and deletion controls, accommodate accessibility needs, measure false-reject rates on representative volunteers, and use behavioral information as an adaptive risk signal alongside passwords, MFA, rate limits, and secure server sessions.

Evaluation checklist

Reliability	Enrollment completes three passes; normal login is evaluated against saved per-signal tolerance.
Innovation	Interaction rhythm becomes a local second factor with a visible decision trail.
Latency	No network call or model download; only fixed-size client-side arithmetic.
Explainability	The dashboard provides confidence, signal status, and a plain-language reason.
Demo readiness	A bot simulator produces a repeatable pre-classification block.

Three-minute demonstration

0:00-0:45 — Enroll. Type the phrase naturally three times. Explain that only timing and movement aggregates are stored locally.

0:45-1:30 — Genuine login. Enter credentials naturally. Show the green decision and individual signal matches.

1:30-2:15 — Impostor rhythm. Ask another person to type deliberately differently; point to the red score and largest deviation.

2:15-3:00 — Bot response. Select the bot simulator. Show the orange pre-classification block and its clear reason.

BioPrint is a demo of behavioral risk assessment, not an authentication system intended to stand alone.